Precision Over Volume: Improving Federal Statistical AI Grounding through Pragmatic Expert Judgment Curation

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Abstract

As federal statistical agencies increasingly deploy Large Language Models (LLMs) to mediate public access to complex data, the challenge of ensuring grounding and fidelity becomes paramount. Traditional Retrieval-Augmented Generation (RAG) approaches rely on large-scale document chunking, which often introduces significant variance and noisy context, termed here as a "stochastic tax." This study evaluates a novel "Pragmatics" approach that replaces high-volume retrieval with low-volume, high-fidelity expert judgment curation. Using 354 pages of U.S. Census Bureau documentation, we demonstrate that 36 curated expert items significantly outperform 311 RAG-retrieved chunks (d = 0.922), with a 16.6 percentage point advantage in interpretation fidelity. We propose a "Sidecar Architecture" for deterministic delivery of these pragmatic items, decoupling expert knowledge from underlying reasoning models. Our findings suggest that for federal statistical interpretation, precision of signal consistently outperforms volume of context.

1 Introduction

Federal statistical agencies are the primary stewards of the nation's data infrastructure. The integrity of these agencies rests *not* only on the accuracy of the estimates they produce but also on the public's ability to interpret those estimates correctly. The Federal Committee on Statistical Methodology (FCSM) framework for data quality (FCSM 20-04) explicitly identifies "fitness for use" as a core dimension of statistical quality [1]. However, as data consumption shifts from static tables to AI-mediated natural language queries, the mechanism for delivering this quality context must evolve.

Current AI architectures primarily address the grounding problem through Retrieval-Augmented Generation (RAG) [2]. While RAG effectively expands a model's knowledge base, it suffers from structural limitations when applied to the nuances of statistical methodology. Embedding-based retrieval is inherently stochastic and often collapses the distance between semantically similar but pragmatically distinct concepts—such as the difference between a margin of error and a coefficient of variation.

This paper introduces the concept of *Pragmatics* in legal and statistical AI: the curated delivery of expert judgment at the point of decision. We argue that the machine learning principle of training data curation extends to inference-time context. By selectively curating the "reasoning traces" that matter most to a senior statistician, agencies can reduce the "stochastic tax"—the accumulation of variance across non-deterministic retrieval and generation pipelines—and provide more reliable guidance to data users.

2 Methods and Contextual Framework

Our study utilized 354 pages of methodology documentation from the U.S. Census Bureau, specifically focusing on complex surveys such as the American Community Survey (ACS) [3].

2.1 The Pragmatic Pipeline

We developed a multi-stage pipeline designed for information distillation rather than accumulation:

1. **Knowledge Graph Extraction:** Source documents were decomposed into 5,233 knowledge graph nodes.

2. **Harvesting:** Candidate items likely to contain expert judgment were identified.

3. **Expert Curation:** A senior statistician reviewed harvested items, selecting only those critical for interpreting results for a given set of queries.

4. **Resulting Pack:** The final "Pragmatic Pack" consisted of only 36 high-fidelity items (a 0.65% retention rate from the initial graph nodes).

2.2 Experimental Design

We compared the Pragmatics approach against two baselines:

- **Volume Baseline (RAG):** A standard retrieval system using 311 chunks from the same source material.

- **Control:** A baseline LLM without additional methodology context.

Fidelity was assessed by a multi-vendor judge panel (Claude, GPT, Gemini) across five dimensions of quality, with a particular focus on uncertainty communication (D3).

3 Results

The empirical results demonstrate a clear inversion of the traditional "more data is better" paradigm.

3.1 The Selectivity Advantage

The central finding is that 36 curated expert judgment items outperform 311 document chunks with a large effect size ($d = 0.922$) and a 16.6% fidelity advantage (Figure 1). This confirms that for high-stakes statistical interpretation, selectivity at inference time follows the same pattern as high-quality training data curation: precision of signal outperforms volume of context.

3.2 Uncertainty Communication (D3)

The most significant impact was observed in the communication of uncertainty. This dimension depends most directly on fitness-for-use judgment. While RAG can retrieve definitions of error margins, the Pragmatics approach delivers the specific judgment that a given margin renders an estimate unreliable for the requested use case. This distinction (D3 dimension) showed an even larger effect ($d = 1.353$) vs. the control.

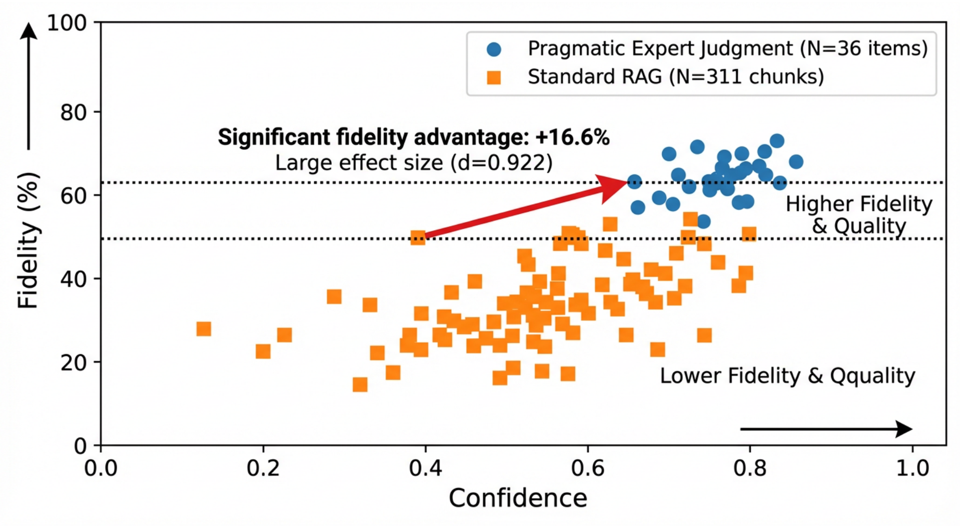

Figure 1: Comparison of Interpretation Fidelity: 36 Curated Items vs. 311 RAG Chunks. Note the significant fidelity advantage despite the 10x reduction in context volume.

4 Discussion: Reducing the Stochastic Tax

Every AI system built on language models pays a "stochastic tax": variance that cannot be eliminated because the underlying mechanism is non-deterministic. However, RAG compounds this tax at two stages: retrieval is stochastic (approximate embedding similarity) and generation is stochastic.

Pragmatics eliminates the first source of compounding. Context retrieval in this framework is deterministic—a graph traversal that returns identical results every time. By fixing the grounding, we isolate variance to the reasoning stage, preventing the "noise on noise" effect that plagues high-volume retrieval systems.

4.1 The Sidecar Architecture

To operationalize these findings, we propose a "Sidecar Architecture" (Figure 2). In this model, statistical expertise is served as a server-side API resource.

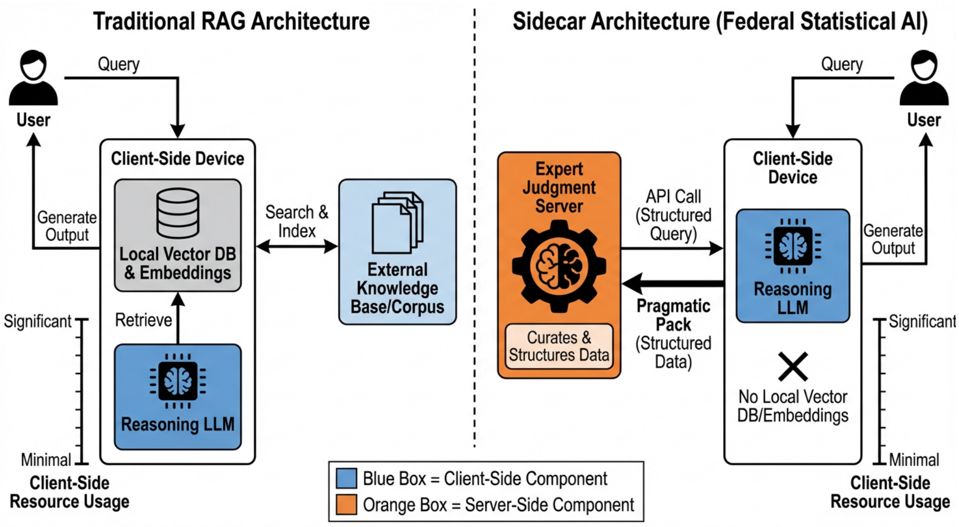

Figure 2: The Sidecar Architecture for deterministic expert judgment delivery. Expert knowledge is decoupled from the reasoning model, allowing for vendor independence and lower client-side infrastructure costs.

Current RAG systems require clients to maintain expensive vector databases and indices. In contrast, the Sidecar Architecture concentrates curation costs at the source (the agency or expert) and distributes benefits through low-cost, structured API calls. This allows agencies to update their methodology guidance centrally, ensuring all users receive the most current expert judgment immediately.

5 Conclusion

The Federal Committee on Statistical Methodology's quality standards—relevance, accuracy, and fitness for use—have been the benchmark for decades. Pragmatics provides the technical bridge to operationalize these standards for the AI era. By treating expert judgment as a machine-readable, deliverable resource rather than just static documentation, federal agencies can ensure that their data is not only accessible but also correctly understood. Precision of signal is the most effective defense against the inherent variance of large language models.

References

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